



Sardar Patel Institute of Technology
Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai: 400058, India
(An Empowered Autonomous College Affiliated to University of Mumbai)
Department of Computer Engineering

**Vehicle Rental Management System (VRMS) –
Work Breakdown Structure and Project Schedule**

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Aim

To plan the Vehicle Rental Management System (VRMS) project by preparing a Work Breakdown Structure (WBS), in which every work package follows the 8/80 rule, and a Project Schedule (Gantt chart with activity list, durations and dependencies) prepared with the GanttProject tool, starting on 20 July 2026 and ending on 12 October 2026, based on the team's Software Requirements Specification (SRS), class diagram and sequence diagrams.

- Work Breakdown Structure – a hierarchical representation of all the work required to complete the project. It divides the project into phases and work packages, which helps in defining scope, assigning responsibilities, estimating effort and planning schedules.
- Project Schedule – organizes the work packages in a logical sequence with estimated durations and dependencies, to determine when each activity starts and finishes and to make effective use of the team's resources.
- 8/80 rule – every work package must need at least 8 and at most 80 hours of effort. A smaller package is too fine to track separately; a larger one is too coarse to estimate and monitor reliably.

Work Breakdown Structure

The WBS divides the complete VRMS project (1.0) into seven major phases and 26 work packages. Every element carries a hierarchical number, so each package can be scheduled, assigned and tracked on its own.

- Level 1 – Project: 1.0 Vehicle Rental Management System, the complete deliverable.
- Level 2 – Major phases: 1.1 Project Planning, 1.2 Requirements Analysis, 1.3 System Design, 1.4 System Development, 1.5 Testing, 1.6 Deployment and 1.7 Maintenance.
- Level 3 – Work packages: the smallest units of work, such as 1.3.2 Class Design or 1.4.3 Booking and Payment.



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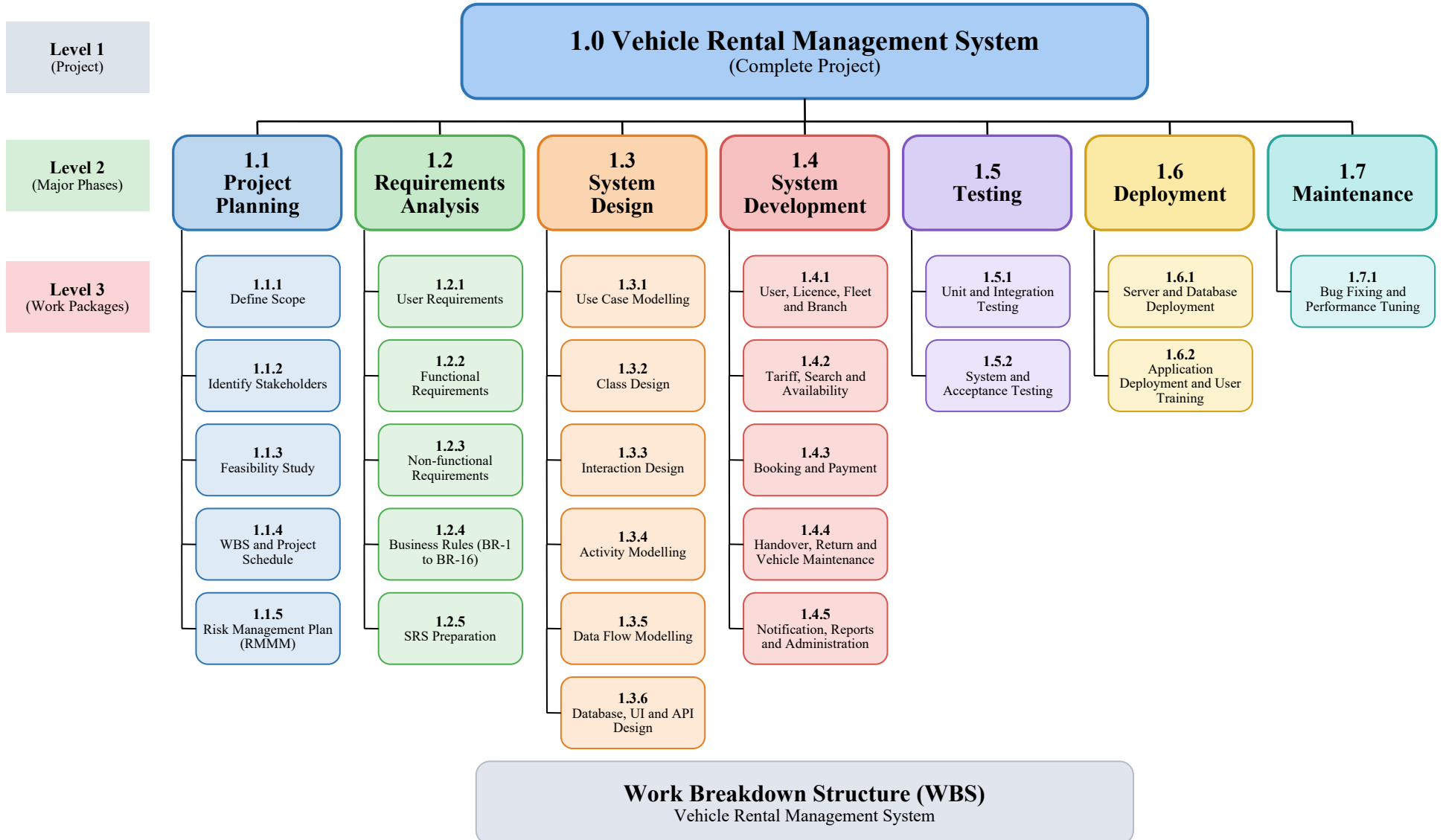


Figure 1: Work Breakdown Structure – Vehicle Rental Management System



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System Development is broken down by the class diagram (Experiment 3) and the sequence diagrams (Experiment 4). Each of its five work packages groups the classes that collaborate in one interaction, so that together they cover all twenty classes of the class diagram. The design work packages correspond one-to-one with the UML and DFD experiments, so each experiment closes a package of the WBS. The table gives every work package with its effort in hours, which shows that the 8/80 rule is satisfied, and the source it is derived from.

Work package	Effort (hours)	Source (SRS / diagram / experiment)
1.1.1 Define Scope	16	SRS 1.2 Project Scope
1.1.2 Identify Stakeholders	8	SRS 2.2 User Classes
1.1.3 Feasibility Study	16	SRS 2.4 Constraints (CO-8)
1.1.4 WBS and Project Schedule	40	Experiment 7
1.1.5 Risk Management Plan (RMMM)	8	Experiment 8
1.2.1 User Requirements	16	SRS 2.2 User Classes
1.2.2 Functional Requirements	24	SRS Section 3
1.2.3 Non-functional Requirements	24	SRS 5.1 to 5.4
1.2.4 Business Rules (BR-1 to BR-16)	24	SRS 5.5
1.2.5 SRS Preparation	24	Experiment 1
1.3.1 Use Case Modelling	48	Experiment 2
1.3.2 Class Design	40	Experiment 3
1.3.3 Interaction Design	80	Experiment 4
1.3.4 Activity Modelling	72	Experiment 5
1.3.5 Data Flow Modelling	40	Experiment 6
1.3.6 Database, UI and API Design	32	SRS CO-4, CO-5, 4.1 User Interfaces
1.4.1 User, Licence, Fleet and Branch	16	User hierarchy, Licence, Vehicle, Car, TwoWheeler, Branch; SRS 3.7
1.4.2 Tariff, Search and Availability	16	Tariff, check availability(); BR-1, BR-6; SRS 3.2
1.4.3 Booking and Payment	16	Booking, Payment subclasses, Invoice; Sequence Diagram 1; SRS 3.1, 3.4
1.4.4 Handover, Return and Vehicle Maintenance	16	ConditionReport, MaintenanceJob; Sequence Diagrams 2 and 3; SRS 3.3, 3.6
1.4.5 Notification, Reports and Administration	8	SRS 3.8, BR-13, audit log
1.5.1 Unit and Integration Testing	8	Experiment 10; Maintainability-2
1.5.2 System and Acceptance Testing	8	Experiment 10; Reliability-1, Usability-1
1.6.1 Server and Database Deployment	8	SRS OE-4, OE-5
1.6.2 Application Deployment and User Training	8	SRS UD-1 to UD-4
1.7.1 Bug Fixing and Performance Tuning	8	SRS PE-1 to PE-6

The smallest package needs 8 hours and the largest, 1.3.3 Interaction Design, needs 80 hours, so every package lies within the 8/80 limits.

Project Schedule

The project runs from 20 July 2026 to 12 October 2026: 63 working days, or 85 calendar days. Activity IDs match the WBS, so every row of the chart can be traced to a work package. The schedule was prepared in the GanttProject tool, and the date of each laboratory experiment is fixed as a milestone, as the table shows.

Experiment	Date	Milestone (work package)
Exp 1 – Software Requirements Specification	Mon 10 Aug 2026	1.2.5 SRS Preparation complete
Exp 2 – Use Case Diagram	Tue 18 Aug 2026	1.3.1 Use Case Modelling complete
Exp 3 – Class Diagram	Tue 25 Aug 2026	1.3.2 Class Design complete



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Exp 4 – Sequence and Collaboration Diagrams	Tue 08 Sep 2026	1.3.3 Interaction Design complete
Exp 5 – Activity Diagram	Mon 21 Sep 2026	1.3.4 Activity Modelling complete
Exp 6 – Data Flow Diagrams	Mon 28 Sep 2026	1.3.5 Data Flow Modelling complete
Exp 7 – WBS and Project Schedule	Mon 05 Oct 2026	1.1.4 WBS and Project Schedule complete
Exp 8 – RMMM Plan	Mon 05 Oct 2026	1.1.5 Risk Management Plan complete
Exp 9 – Module Implementation	Tue 06 Oct 2026	1.4 System Development begins
Exp 10 – Testing and Evaluation	Mon 12 Oct 2026	1.5 Testing complete; project end



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Project Schedule – Vehicle Rental Management System
(Gantt Chart with Activity List, Duration and Dependencies)

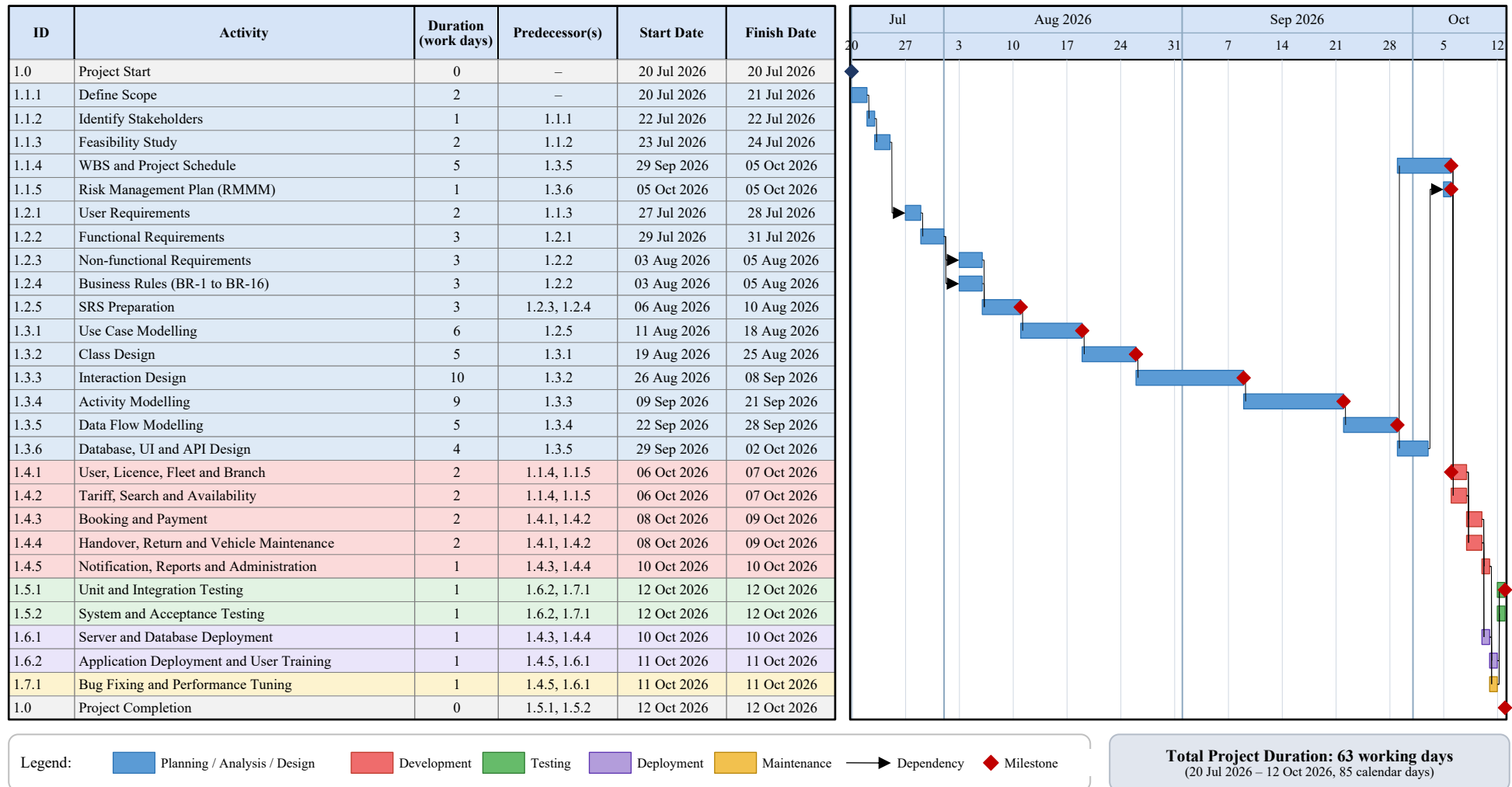


Figure 2: Project Schedule – Vehicle Rental Management System (Gantt chart)



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The schedule is built on these assumptions:

- Durations are in working days (Monday to Friday); public holidays are not deducted. Saturday 10 and Sunday 11 October are treated as working days, so that implementation and deployment finish before the evaluation on 12 October.
- Each team member works 8 hours a day, so the effort of a work package is its duration multiplied by 8 hours.
- Every dependency is finish-to-start: an activity begins on the first working day after its last predecessor finishes.
- The team has two members, so no more than two activities run in parallel.
- The date of each experiment is fixed, and the work package it represents finishes on that date.
- Because implementation begins with Experiment 9 and testing is Experiment 10, the system is deployed on 10 and 11 October and the final tests on 12 October are carried out on the deployed system.

Phase	Start	Finish	Working days
Project Planning – initiation (1.1.1 to 1.1.3)	20 Jul 2026	24 Jul 2026	5
Requirements Analysis (1.2)	27 Jul 2026	10 Aug 2026	11
System Design (1.3)	11 Aug 2026	02 Oct 2026	39
Project Planning – WBS, schedule and RMMM (1.1.4, 1.1.5)	29 Sep 2026	05 Oct 2026	5
System Development (1.4)	06 Oct 2026	10 Oct 2026	5
Deployment (1.6)	10 Oct 2026	11 Oct 2026	2
Maintenance (1.7)	11 Oct 2026	11 Oct 2026	1
Testing (1.5)	12 Oct 2026	12 Oct 2026	1

The order of the development packages follows the sequence diagrams, so each module is built after the modules whose classes it calls:

- 1.4.1 User, Licence, Fleet and Branch (Rohan) and 1.4.2 Tariff, Search and Availability (Nidhi) come first and run side by side, because Sequence Diagram 1 calls `is_valid()` on Licence and `check_availability()` on Vehicle before a booking exists.
- 1.4.3 Booking and Payment (Rohan) needs both, since `create_booking()` requires a verified licence, an available vehicle and a tariff, and `authorize()` and `generate_invoice()` follow it. 1.4.4 Handover, Return and Vehicle Maintenance (Nidhi) runs beside it and follows Sequence Diagrams 2 and 3.
- 1.4.5 Notification, Reports and Administration (Rohan) comes last because it uses events from every other module, while 1.6.1 Server and Database Deployment (Nidhi) runs beside it.

Because every experiment date is a fixed milestone, the schedule has no slack. The critical path is 1.1.1 → 1.1.2 → 1.1.3 → 1.2.1 → 1.2.2 → 1.2.3 / 1.2.4 → 1.2.5 → 1.3.1 → 1.3.2 → 1.3.3 → 1.3.4 → 1.3.5 → 1.1.4 / 1.3.6 → 1.1.5 → 1.4.1 / 1.4.2 → 1.4.3 / 1.4.4 → 1.4.5 / 1.6.1 → 1.6.2 / 1.7.1 → 1.5.1 / 1.5.2. Requirements Analysis ends on 10 August 2026, the date of the SRS.



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Conclusion

The Work Breakdown Structure and Project Schedule for the Vehicle Rental Management System were prepared from the team's SRS, class diagram and sequence diagrams. The WBS splits the project into seven phases and 26 work packages. Every package needs between 8 and 80 hours of effort, so the 8/80 rule is satisfied, and the five development packages together cover all twenty classes of the class diagram.

The schedule, prepared with the GanttProject tool, sequences those packages with explicit durations and finish-to-start dependencies. Starting on 20 July 2026, it completes on 12 October 2026 after 63 working days (85 calendar days). The SRS is finished on 10 August, the design experiments between 18 August and 28 September, the WBS, schedule and risk plan on 5 October, implementation begins on 6 October and testing is completed on 12 October. Together, the two diagrams give the team a clear basis for assigning work, tracking progress and spotting delays early.